Lagrange Multipliers: A Simple Introduction

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Why do we need Lagrange multipliers?

Suppose that we want to maximise a function

$$f(x, y).$$

If there are no restrictions on the values of x and y, we can differentiate with respect to each variable and solve

$$\frac{\partial f}{\partial x} = 0, \qquad \frac{\partial f}{\partial y} = 0.$$

However, sometimes we are **not free to choose x and y independently**.

For example, suppose that we want to maximise $f(x, y)$ subject to the restriction

$$x + y = 10.$$

This means that we cannot choose any values of x and y that we want.

If we choose $x = 4$, then we must have

$$y = 6.$$

If we choose $x = 7$, then we must have

$$y = 3.$$

So the values of x and y must always satisfy the constraint.

The main idea

A **Lagrange multiplier** is simply a mathematical tool that allows us to maximise or minimise a function while keeping a constraint satisfied.

Suppose that we want to maximise

$$f(x, y)$$

subject to

$$g(x, y) = c.$$

Instead of treating the maximisation and the constraint separately, we combine them into a new function called the **Lagrangian**:

$$\mathcal{L}(x, y, \lambda) = f(x, y) + \lambda\{c - g(x, y)\}.$$

Here, λ is called the **Lagrange multiplier**.

The role of λ is to allow us to include the constraint directly in the optimisation problem.

We then differentiate the Lagrangian with respect to all the unknown quantities:

$$\frac{\partial \mathcal{L}}{\partial x} = 0,$$

$$\frac{\partial \mathcal{L}}{\partial y} = 0,$$

and

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 0.$$

The last equation simply gives us the original constraint.

Why does differentiating with respect to λ recover the constraint?

From

$$\mathcal{L}(x, y, \lambda) = f(x, y) + \lambda\{c - g(x, y)\},$$

we have

$$\frac{\partial \mathcal{L}}{\partial \lambda} = c - g(x, y).$$

Therefore,

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 0$$

implies

$$c - g(x, y) = 0,$$

or equivalently,

$$g(x, y) = c.$$

So the Lagrange multiplier method forces the solution to satisfy the constraint.

A simple example

Suppose that we want to maximise

$$f(x, y) = xy$$

subject to

$$x + y = 10.$$

We know that x and y cannot be chosen independently because they must add to 10.

We construct the Lagrangian:

$$\mathcal{L}(x, y, \lambda) = xy + \lambda(10 - x - y).$$

Now differentiate with respect to x:

$$\frac{\partial \mathcal{L}}{\partial x} = y - \lambda.$$

Setting this equal to zero gives

$$y - \lambda = 0,$$

so

$$y = \lambda.$$

Similarly,

$$\frac{\partial \mathcal{L}}{\partial y} = x - \lambda.$$

Therefore,

$$x = \lambda.$$

Hence,

$$x = y.$$

Finally, differentiate with respect to λ:

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 10 - x - y.$$

Setting this equal to zero gives

$$x + y = 10.$$

Since $x = y$,

$$2x = 10,$$

and therefore

$$x = y = 5.$$

Thus, the maximum occurs at

$$(x, y) = (5, 5).$$

What should we remember?

The important idea is:

> Lagrange multipliers are useful when we want to maximise or minimise something, but the parameters must satisfy a restriction.

The steps are:

1. Write down the function that we want to maximise.
2. Write down the constraint.
3. Add the constraint to the function using a Lagrange multiplier λ.
4. Differentiate with respect to all the unknowns.
5. Set the derivatives equal to zero.
6. Solve the resulting equations together.

Why will this be useful for the EM algorithm?

In mixture models, the mixing probabilities satisfy

$$\pi_1 + \pi_2 + \cdots + \pi_K = 1.$$

Therefore, when we maximise with respect to

$$\pi_1, \ldots, \pi_K,$$

we cannot treat them as completely independent parameters.

For example, with $K = 3$,

$$\pi_1 + \pi_2 + \pi_3 = 1.$$

If π_1 and π_2 are fixed, then π_3 is automatically determined:

$$\pi_3 = 1 - \pi_1 - \pi_2.$$

This is exactly the type of situation for which Lagrange multipliers are useful.

Later, in the M-step of the EM algorithm, we will want to maximise a function involving

$$\pi_1, \pi_2, \pi_3$$

while still requiring

$$\pi_1 + \pi_2 + \pi_3 = 1.$$

We will therefore construct a Lagrangian of the form

$$\mathcal{L} = Q(\pi_1, \pi_2, \pi_3) + \lambda(1 - \pi_1 - \pi_2 - \pi_3),$$

and then differentiate with respect to

$$\pi_1, \qquad \pi_2, \qquad \pi_3,$$

and

$$\lambda.$$

The purpose of the Lagrange multiplier is simply to make sure that, while we maximise Q, the mixing probabilities still add to one.